

BASE BELGRANO II, Antarctica

WMO: 890340

Lat: 77.874S

Lon: 34.626W

Elev: 256

StdP: 98.29

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) -34.6	(c) -32.1	(d) -43.8	(e) 0.1	(f) -31.4	(g) -40.9	(h) 0.1	(i) -30.8	(j) 26.7	(k) -13.5	(l) 25.7	(m) -15.6	(n) 3.0	(o) 140	(p) 1.229

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 12	(b) 5.5	(c) 3.9	(d) 0.9	(e) 2.6	(f) 0.2	(g) 1.5	(h) -0.5	(i) 1.4	(j) 3.0	(k) 0.5	(l) 2.2	(m) -0.3	(n) 1.3	(o) 3.0	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
0.0	3.9	1.6	-0.9	3.6	0.7	-1.9	3.3	0.0	12.4	3.1	11.0	2.3	9.5	1.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	20.5	15.2	-37.6	6.2	2.1	0.5	-39.1	6.5	-40.4	6.8	-41.5	7.1	-43.0	7.5
			-38.2	2.9	3.3	1.7	-40.6	4.1	-42.5	5.1	-44.3	6.0	-46.7	7.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-13.2	-2.7	-6.5	-11.9	-16.5	-17.5	-19.0	-21.0	-20.5	-18.7	-14.1	-6.9	-2.5	(6)
	DBStd	8.17	2.56	3.67	4.99	5.97	5.45	5.06	5.43	6.10	5.86	4.84	3.47	2.77	(7)
	HDD10.0	8459	394	463	680	796	852	869	961	944	861	746	506	388	(8)
	HDD18.3	11501	652	696	938	1046	1110	1119	1219	1203	1111	1004	756	646	(9)
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(10)
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
	WSAvg	5.4	4.0	5.3	5.4	5.6	5.8	5.6	6.2	5.6	6.0	5.5	5.4	4.7	(14)
(15) Precipitation	PrecAvg	272	14	15	17	19	20	26	27	31	33	26	28	16	(15)
	PrecMax	420	39	42	41	39	39	94	63	61	78	92	93	49	(16)
	PrecMin	165	3	1	2	1	1	7	3	2	5	1	3	1	(17)
	PrecStd	57	9	10	9	11	12	20	16	15	20	20	22	12	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.0	2.8	-1.8	-2.0	-3.3	-5.0	-8.7	-4.8	-5.5	-1.3	3.4	5.6	(19)
		MCWB	1.3	0.5	-3.0	-2.7	-4.1	-6.2	-9.6	-6.2	-6.2	-3.6	0.3	2.0	(20)
	2%	DB	3.6	1.1	-3.2	-4.6	-6.0	-7.4	-10.1	-8.3	-7.4	-4.0	0.8	4.3	(21)
		MCWB	0.7	-0.5	-4.4	-5.2	-6.8	-8.4	-10.9	-9.1	-8.4	-5.3	-1.0	1.2	(22)
	5%	DB	2.5	-0.3	-4.3	-6.2	-8.5	-9.8	-11.7	-10.2	-9.1	-5.8	-0.7	3.0	(23)
		MCWB	0.1	-1.7	-5.3	-6.9	-9.1	-10.6	-12.6	-10.8	-10.2	-7.0	-2.1	0.4	(24)
	10%	DB	1.4	-1.6	-5.2	-8.2	-10.4	-11.8	-13.1	-12.2	-10.8	-7.5	-2.0	1.8	(25)
		MCWB	-0.5	-2.9	-6.2	-8.8	-11.2	-12.6	-13.8	-12.9	-11.7	-8.5	-3.3	-0.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.6	1.0	-2.8	-2.8	-4.0	-6.0	-9.4	-5.9	-6.0	-3.1	1.2	2.8	(27)
		MCDB	3.7	2.2	-2.3	-2.3	-3.4	-5.2	-8.8	-4.9	-5.6	-2.0	1.9	4.6	(28)
	2%	WB	1.2	-0.2	-4.1	-5.1	-6.6	-8.4	-10.9	-9.0	-8.2	-5.2	-0.6	1.6	(29)
		MCDB	2.8	0.8	-3.4	-4.7	-5.9	-7.5	-10.2	-8.4	-7.5	-4.3	0.6	3.5	(30)
	5%	WB	0.4	-1.4	-5.2	-6.8	-9.1	-10.5	-12.5	-10.8	-10.0	-6.9	-1.9	0.7	(31)
		MCDB	2.1	-0.5	-4.4	-6.3	-8.6	-9.8	-11.8	-10.2	-9.2	-6.0	-1.0	2.5	(32)
	10%	WB	-0.4	-2.7	-6.2	-8.7	-11.1	-12.6	-13.8	-12.9	-11.7	-8.4	-3.1	-0.1	(33)
		MCDB	1.2	-1.7	-5.3	-8.1	-10.4	-11.8	-13.0	-12.3	-10.7	-7.6	-2.2	1.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	5.2	5.9	5.3	5.2	5.3	5.5	5.5	5.6	5.8	6.8	6.6	5.5	(35)
		MCDBR	7.0	7.1	5.0	4.8	5.9	5.9	6.2	6.6	6.4	7.8	8.1	7.5	(36)
	5% WB	MCWBR	5.6	6.7	4.7	4.7	5.6	5.8	5.8	6.3	5.9	7.3	7.2	5.9	(37)
		MCWBR	6.7	7.1	4.8	4.6	5.7	5.8	6.2	6.5	5.8	7.5	7.9	7.1	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.240	0.222	0.181	0.116	N/A	N/A	N/A	0.118	0.175	0.201	0.219	0.238	(40)
		taud	2.252	2.257	2.187	1.904	N/A	N/A	N/A	1.877	2.126	2.218	2.250	2.243	(41)
	Ebn at Noon	Ebn at Noon	957	890	751	106	0	0	0	80	736	917	983	984	(42)
		Edh at Noon	98	76	46	3	0	0	0	3	47	78	97	105	(43)
(44) All-Sky Solar Radiation	RadAvg	7.85	4.71	1.59	0.13	0.00	0.00	0.00	0.00	0.02	0.82	3.59	7.31	9.27	(44)
	RadStd	0.32	0.22	0.08	0.01	0.00	0.00	0.00	0.00	0.00	0.04	0.13	0.20	0.27	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page